CASE STUDY

Intelligent Node Provisioning with Karpenter on Amazon EKS

Migrating from Kubernetes Cluster Autoscaler to Karpenter to improve scalability, flexibility, and cost efficiency on Amazon EKS.

Managing resource allocation and autoscaling is critical in cloud-native environments. OPT IT Technologies helped modernize Amazon EKS by migrating from Kubernetes Cluster Autoscaler to Karpenter to overcome scaling limitations and improve infrastructure efficiency.

Challenges

- Cluster Autoscaler required pre-defined node groups with fixed instance types, limiting flexibility in workload placement.
- Slow node provisioning during traffic spikes, leading to pods stuck in pending state for several minutes.
- Over-provisioned nodes due to rigid instance type selection, resulting in underutilized compute and higher EC2 costs.
- No awareness of workload-specific requirements - Cluster Autoscaler could not match instance types to individual pod resource requests.

Solution

- **Phased migration to Karpenter**
- **Configured Karpenter NodePools**
- **Performed scaling and performance testing**
- **Enabled monitoring with Prometheus & Grafana**

Results

- **Achieved intelligent autoscaling**
- **Enhanced workload scalability**
- **Improved resource utilization**
- **Increased cost efficiency**
- **Reduced operational costs**

Value Delivered

- **Delivered a seamless migration to karpenter**
- **Intelligent autoscaling**
- **Improved infrastructure efficiency**
- **Optimized kubernetes resource management**

Conclusion

OPT IT Technologies successfully modernized Amazon EKS by implementing Karpenter, enabling intelligent autoscaling, improved scalability, better resource utilization, and greater cost efficiency while ensuring a seamless migration experience.

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